

Week 8: Modeling Volatility (ARCH)

Geometric series

The geometric series is

$$\sum_{j=0}^{\infty} a^j = 1 + a + a^2 + \dots = \frac{1}{1-a},$$

provided $|a| < 1$.

$$a = 0.5$$

$$\downarrow$$
$$\frac{1}{1-0.5}$$

h-step ahead forecast

A one-step ahead forecast

$c(1)$

$$\mathbb{E}(y_{t+1}|\mathcal{F}_t) = c + \phi_1 y_t$$

A two-step ahead forecast is

$h=2$

$$\mathbb{E}(y_{t+2}|\mathcal{F}_t) = c(1 + \phi_1) + \phi_1^2 y_t$$

A three-step ahead forecast

$h=3$

$$\mathbb{E}(y_{t+3}|\mathcal{F}_t) = c(1 + \phi_1 + \phi_1^2) + \phi_1^3 y_t.$$

What about more than one-step ahead?

Taking **many steps ahead** to the limit we have

$$\begin{aligned} \lim_{h \rightarrow \infty} \mathbb{E}(y_{t+h} | \mathcal{F}_t) &= \lim_{h \rightarrow \infty} c(1 + \phi_1 + \dots + \phi_1^{h-1}) + \lim_{h \rightarrow \infty} \phi_1^h y_t \\ &= \underbrace{c \sum_{h=0}^{\infty} \phi_1^h}_{\text{geometric series}} + \left(\lim_{h \rightarrow \infty} \phi_1^h \right) y_t \\ &= \frac{c}{1 - \phi_1} + 0. \\ &= E(y_t) \end{aligned}$$

Handwritten notes:

- $1 + a + a^2 + \dots$ (above the first equation)
- far into the future (with an arrow pointing down to $h \rightarrow \infty$)
- $\phi_1 = 0.5$
- $0.5^{1000} \approx 0$
- $\frac{1}{1 - \phi_1}$ (boxed)
- slide 2 (below the boxed fraction)

What about more than one-step ahead?

For stationary models, this many steps ahead conditional expectation converges to the **unconditional mean**

$$\lim_{h \rightarrow \infty} \mathbb{E}(y_{t+h} | \mathcal{F}_t) = \frac{c}{1 - \phi_1} = \mathbb{E}(y_t).$$

An **alternative** way to derive this **unconditional** ('long-run') mean.
Take unconditional expectation and assume 'stationarity':

$$\begin{aligned}\mathbb{E}[y_t] &= \mathbb{E}[c + \phi_1 y_{t-1} + \varepsilon_t] \\ &= c + \phi_1 \mathbb{E}[y_{t-1}] + \mathbb{E}[\varepsilon_t] \\ &= c + \phi_1 \mathbb{E}[y_{t-1}] + 0\end{aligned}$$

AR(1)

ε_t is mds

↓

LIE $\mathbb{E}(\varepsilon_t) = 0$

Stationarity in $y_t \Rightarrow \mathbb{E}[y_t] = \mathbb{E}[y_{t-1}] \Rightarrow$ for all $t \Rightarrow$

$$\mathbb{E}[y_t] = c + \phi_1 \mathbb{E}[y_t]$$

$$\mathbb{E}[y_t] = \frac{c}{1 - \phi_1}$$

The unconditional ('long-run') variance AR(1)

$$\begin{aligned}\mathbb{V}(y_t) &= \mathbb{V}(c + \phi_1 y_{t-1} + \varepsilon_t) \\ &= \mathbb{V}(\phi_1 y_{t-1}) + \mathbb{V}(\varepsilon_t) \\ &= \phi_1^2 \mathbb{V}(y_{t-1}) + \sigma_\varepsilon^2\end{aligned}$$

$$\begin{aligned}\text{var}(\varepsilon_t | \mathcal{F}_{t-1}) \\ &= \text{var}(\varepsilon_t) \\ &= \sigma_\varepsilon^2\end{aligned}$$

Stationarity y_t : $\mathbb{V}(y_t) = \mathbb{V}(y_{t-1})$

$$\begin{aligned}\mathbb{V}(y_t) &= \phi_1^2 \mathbb{V}(y_t) + \sigma_\varepsilon^2 \\ &= \frac{\sigma_\varepsilon^2}{1 - \phi_1^2} \Rightarrow \text{a number}\end{aligned}$$

Note that $\text{Cov}(y_{t-1}, \varepsilon_t) = 0$.

$$\text{Cov}(x, \varepsilon_t) = 0 \quad \text{Intro Econometrics}$$

Conditional variance of AR(1) processes

A time series y_t is an **AR(1) process** if

Week 7

$$y_t = c + \phi_1 y_{t-1} + \varepsilon_t$$

where ε_t is an MDS and $\mathbb{V}(\varepsilon_t | \mathcal{F}_{t-1}) = \mathbb{V}(\varepsilon_t) = \sigma_\varepsilon^2$ *a number*

Recall (Week 7) that the **conditional variance** of y_t is

$$\mathbb{V}(y_t | \mathcal{F}_{t-1}) = \sigma_\varepsilon^2. \text{ a number.}$$

time invariant.

- ▶ The conditional variance differs from the unconditional variance...

- ▶ since $c + \phi_1 y_{t-1}$ is known at time $t - 1$.

- ▶ we have

$$\mathbb{V}(y_t | \mathcal{F}_{t-1}) < \mathbb{V}(y_t)$$

$$\sigma_\varepsilon^2 < \frac{\sigma_\varepsilon^2}{1 - \phi_1}$$

↗ not using any
past info
↓
unconditional.

- ▶ This aligns with the intuition that the use of information reduces uncertainty.

↳ smaller variance.

AR(1) processes

$$Y_t = c + \phi_1 Y_{t-1} + \varepsilon_t$$

For an AR(1) model, the autocorrelations

$$\text{Corr}(y_t, y_{t-k}) = \gamma(k) = \phi_1^k, \quad \text{for } k = 1, 2, 3, \dots$$

$$|\phi_1| < 1$$

So $\gamma(k) \rightarrow 0$ as $k \rightarrow \infty$.

e.g. $\phi_1 = 0.5$

$$k=1 \quad \gamma(1) = \phi_1 \quad \Rightarrow \quad \text{corr}(Y_t, Y_{t-1})$$

$$k=2 \quad \gamma(2) = \phi_1^2 \quad \Rightarrow \quad \text{corr}(Y_t, Y_{t-2})$$

$$k=3 \quad \gamma(3) = \phi_1^3$$
$$\vdots$$

AR(1) processes - unconditional moments

Recall that for an AR(1) model

$$\gamma(k) = \phi_1^k, \quad \text{for } k = 1, 2, 3, \dots$$

slide 12/13

This decline in autocorrelations will be quite rapid if $|\phi_1|$ is small (say $\phi_1 \leq 0.2$), but will remain strong if $|\phi_1|$ is quite large (say $\phi_1 \geq 0.9$).

slide 16/17

$|\phi_1|$ is a measure of ‘persistence’.

Notice that we really do need $|\phi_1| < 1$ for the correlation to decline as $k \rightarrow \infty$.

Simulated AR(1) with $\phi_1 = 0.2$

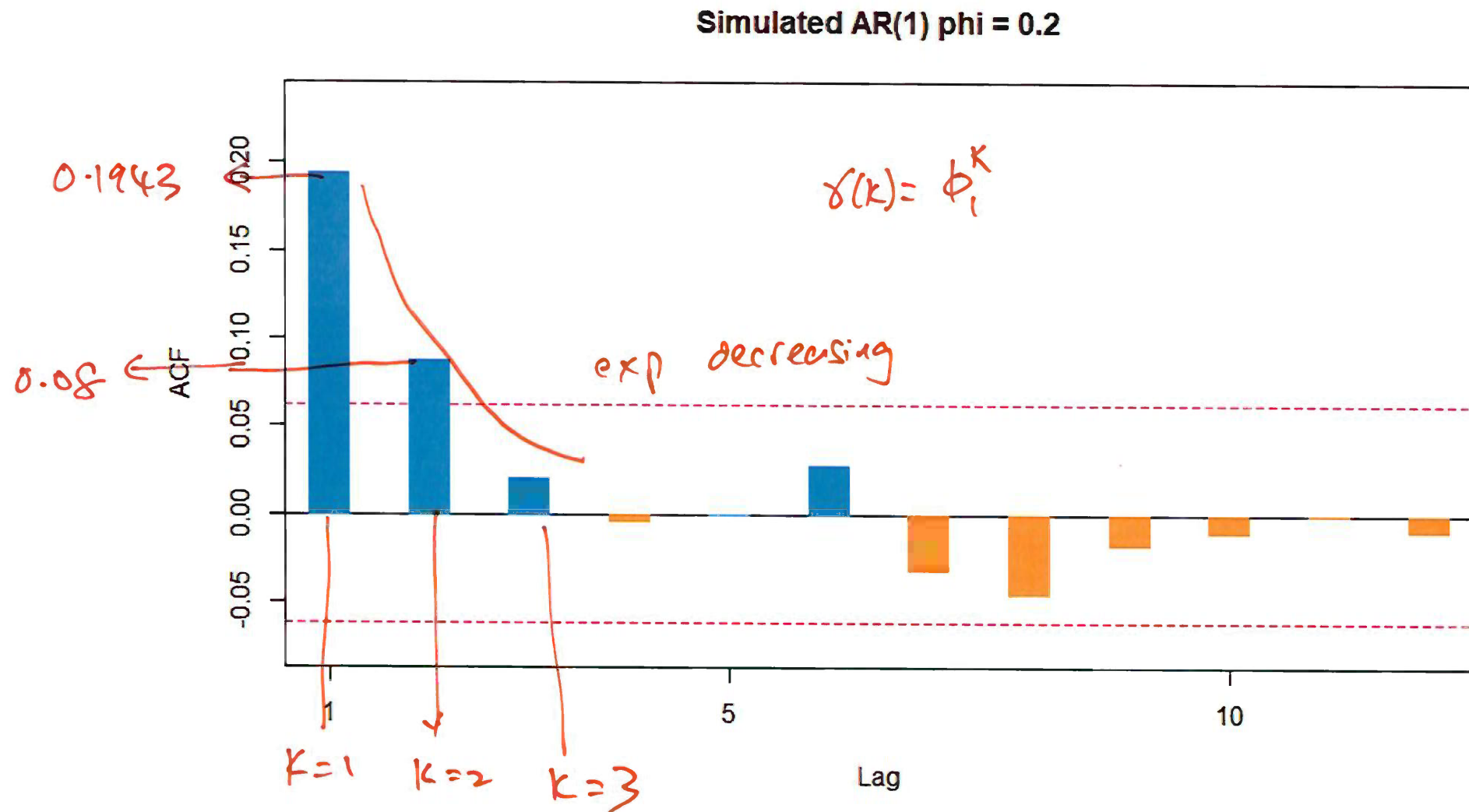
↓
true value

```
> correlogram(y,max.lag = 12, main.acf = "Simulated AR(1) f
```

Lag	AC	Q	Prob
k = 1	0.1943	37.8514	7.634e-10
2	0.0870	45.4581	1.346e-10
3	0.0202	45.8692	6.047e-10
4	-0.0038	45.8834	2.604e-09
5	0.0000	45.8834	9.592e-09
6	0.0280	46.6757	2.172e-08
7	-0.0318	47.6990	4.077e-08
8	-0.0453	49.7712	4.522e-08
9	-0.0179	50.0936	1.034e-07
10	-0.0101	50.1964	2.456e-07
11	-0.0002	50.1964	5.771e-07
12	-0.0094	50.2852	1.245e-06

$\hat{\gamma}(1)$
= 0.2
= ϕ_1

Simulated AR(1) with $\phi_1 = 0.2$



Simulated AR(1) with $\phi_1 = 0.5$

```
> correlogram(y,max.lag = 12, main.acf = "Simulated AR(1) ph
```

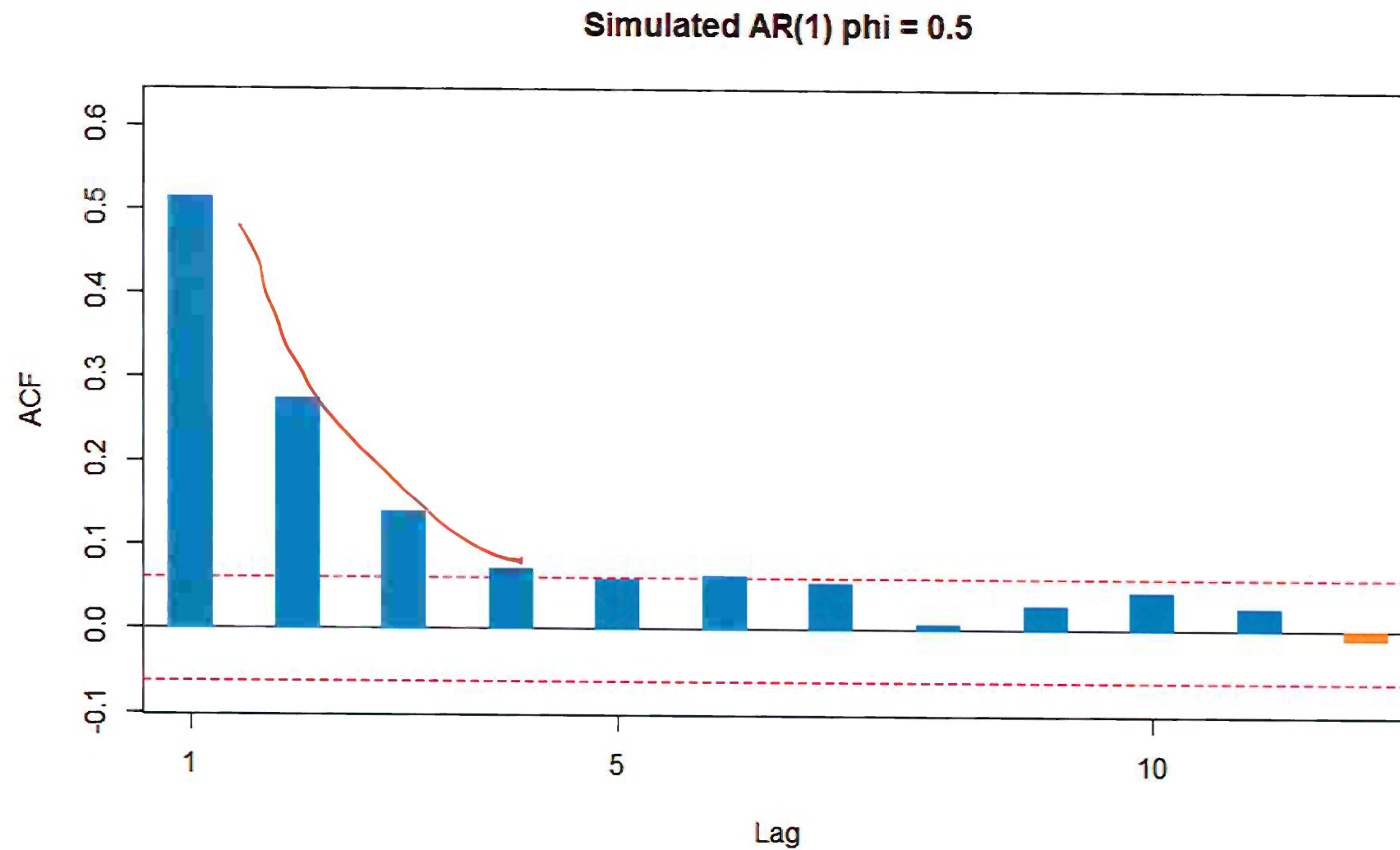
Lag	AC	Q	Prob
$k=1$ 1	0. <u>5131</u>	264.0534	2.244e-59
$k=2$ 2	0. <u>2753</u>	340.1260	1.389e-74
3	0. <u>1376</u>	359.1671	1.544e-77
4	0.0718	364.3588	1.391e-77
5	0.0601	367.9913	2.338e-77
6	0.0626	371.9450	2.990e-77
7	0.0549	374.9853	5.491e-77
8	0.0077	375.0453	4.054e-76
9	0.0292	375.9062	1.882e-75
10	0.0459	378.0346	4.424e-75
11	0.0262	378.7320	1.990e-74
12	-0.0090	378.8144	1.150e-73

$$k=1 \quad \phi_1 = 0.5$$

$$k=2 \quad \phi_1^2 = 0.25$$

$$k=3 \quad \phi_1^3 = 0.125$$

Simulated AR(1) with $\phi_1 = 0.5$



Simulated AR(1) with $\phi_1 = 0.9$

```
> correlogram(y,max.lag = 12, main.acf = "Simulated AR(1) p
```

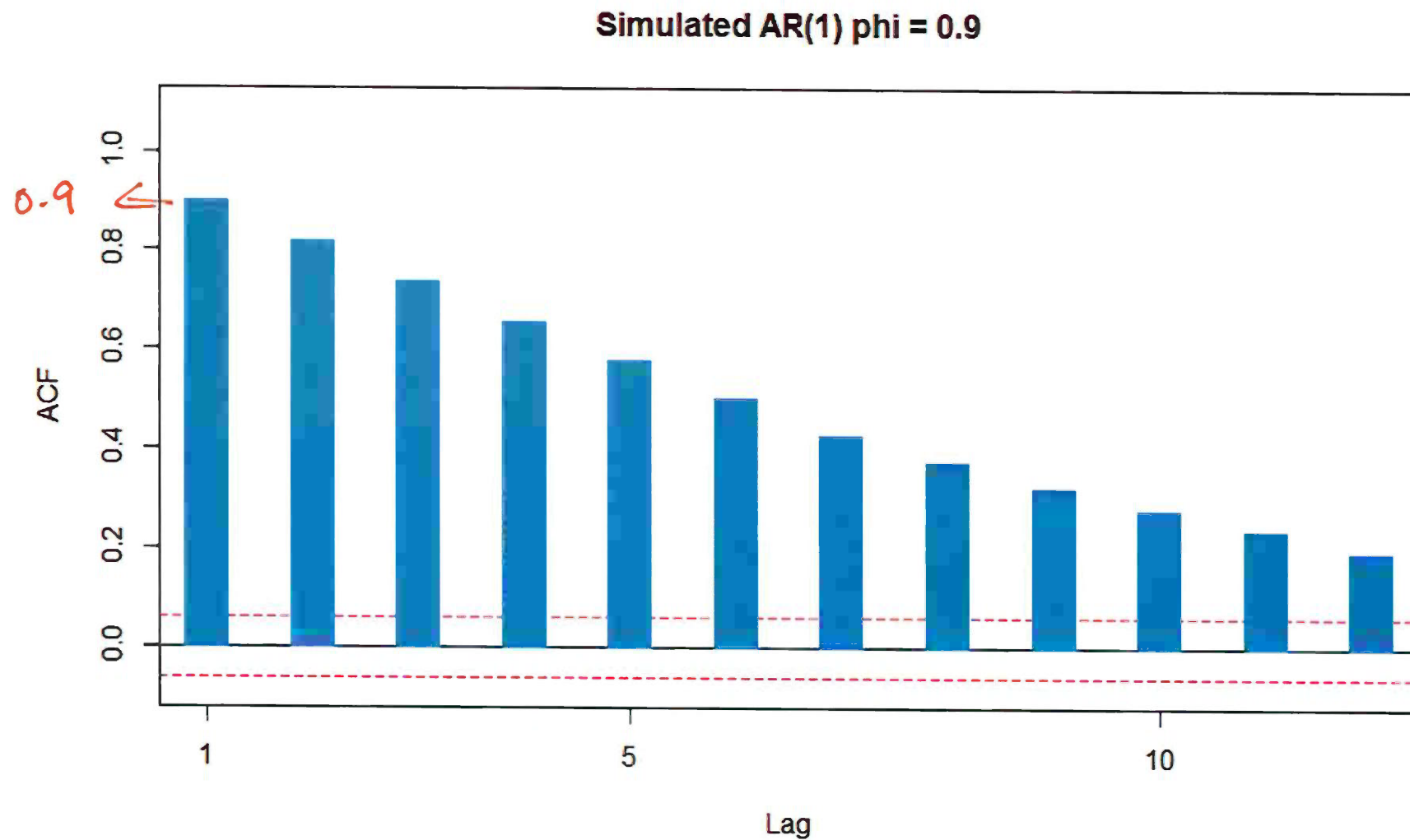
Lag	AC	Q	Prob
1	0.9006	813.4484	6.429000e-179
2	0.8183	1485.7031	2.470328e-323
3	0.7368	2031.3312	0.000000e+00
4	0.6569	2465.4782	0.000000e+00
5	0.5768	2800.4691	0.000000e+00
6	0.5022	3054.6581	0.000000e+00
7	0.4277	3239.2210	0.000000e+00
8	0.3739	3380.4568	0.000000e+00
9	0.3228	3485.8237	0.000000e+00
10	0.2775	3563.7788	0.000000e+00
11	0.2372	3620.7902	0.000000e+00
12	0.1939	3658.9100	0.000000e+00

$$\phi_1 = 0.9 \quad k=1$$

$$\phi_1^2 = 0.81 \quad k=2$$

$$\phi_1^3 = 0.729 \quad k=3$$

Simulated AR(1) with $\phi_1 = 0.9$



A time series y_t is an **AR(p) process** if

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \varepsilon_t$$

Unconditional mean

AR(1)

$$\mathbb{E}[y_t] = \frac{c}{1 - \phi_1 - \dots - \phi_p}.$$

$$\frac{c}{1 - \phi_1}$$

Unconditional variance

$$\mathbb{V}(y_t) = \frac{\sigma_\varepsilon^2}{1 - \sum_{i=1}^p \phi_i \rho_i}.$$

$$\frac{\sigma_\varepsilon^2}{1 - \phi_1^2}$$

AR(p) processes

- ▶ The calculation of $\mathbb{V}(y_t) = \sigma_y^2$ for $p > 1$ is complicated because of covariances between y_{t-1}, y_{t-2} etc.
- ▶ The $\gamma(k)$ formulae are complicated but the autocorrelogram will show that $\gamma(k) \rightarrow 0$ as k increases.
- ▶ The “speed” that $\gamma(k) \rightarrow 0$ and persistence will depend on $\sum_{i=1}^p \phi_i$.
- ▶ $\sum_{i=1}^p \phi_i$ is a persistence measure.

Forecasting using AR(p) models

- ▶ One-step ahead point forecast from $AR(p)$ has conditional mean

$$\mathbb{E}[y_{t+1}|\mathcal{F}_t] = c + \phi_1 y_t + \phi_2 y_{t-1} + \cdots + \phi_p y_{t-p+1} + 0.$$

- ▶ Estimate coefficients using OLS (based on an effective sample from $p + 1, \dots, T$)

$$\underbrace{\hat{\mathbb{E}}[y_{t+1}|\mathcal{F}_t]}_{\text{tmr}} = \hat{c} + \hat{\phi}_1 y_t + \hat{\phi}_2 y_{t-1} + \cdots + \hat{\phi}_p y_{t-(p-1)}.$$

- ▶ Two-steps ahead forecasts can be constructed as

$$\hat{\mathbb{E}}[y_{t+2}|\mathcal{F}_t] = \hat{c} + \hat{\phi}_1 \underbrace{\hat{\mathbb{E}}[y_{t+1}|\mathcal{F}_t]}_{\text{tmr}} + \hat{\phi}_2 y_t + \cdots + \hat{\phi}_p y_{t-(p-2)}.$$

|
today

Comments on the use of AR models for forecasting

- ▶ Forecasts based on well-specified AR models are consistent (i.e. bias is negligible in large samples).
- ▶ But we need to choose our lag length p carefully!
- ▶ Include enough lags to ensure that there is **no left over serial correlation in the residuals**.
- ▶ Then we can interpret ε_t as ‘news’ (unpredictable market forces).

AR(1) $y_t = \phi_1 y_{t-1} + \varepsilon_t$

True model AR(2)

↓

not be white noise

↓

capture y_{t-2}

serially correlated

IF ε_t is white noise AR(1) is correctly specified

Basic Volatility Models

Basic Volatility Models

- Our model for returns is

$$r_t = E[r_t | \mathcal{F}_{t-1}] + \varepsilon_t,$$

→ e.g. AR(1) $c + \phi_1 r_{t-1}$

with $\varepsilon_t \sim \text{WN}(0, \sigma_\varepsilon^2)$.

- Up to now we have focused on modelling the conditional mean return with

$$E[r_t | \mathcal{F}_{t-1}] = c + \phi_1 r_t + \phi_2 r_{t-1} + \dots + \phi_p r_{t-(p-1)},$$

keeping the conditional variance constant, i.e.

$$\mathbb{V}(\varepsilon_t | \mathcal{F}_{t-1}) = \sigma_\varepsilon^2.$$

not realistic
↓
a number ⇒ time invariant.

- Now we will look at modeling the conditional variance as

$$\mathbb{V}(\varepsilon_t | \mathcal{F}_{t-1}) = g(\mathcal{F}_{t-1}).$$

$\mathcal{F}_{t-1} = (r_{t-1}, r_{t-2}, \dots)$

today "t"

now it depends on "t"

past info.

Mean Returns

1. Need a means of modeling the risk of large deviations of returns.
2. Need a way to model volatility.
3. Fortunately, very good frameworks exist to capture this behavior.

want volatility to change over time.

What is volatility?

Volatility is a characterization of risk associated with an asset.

If volatility is high then asset prices are changing more rapidly than when volatility is low.

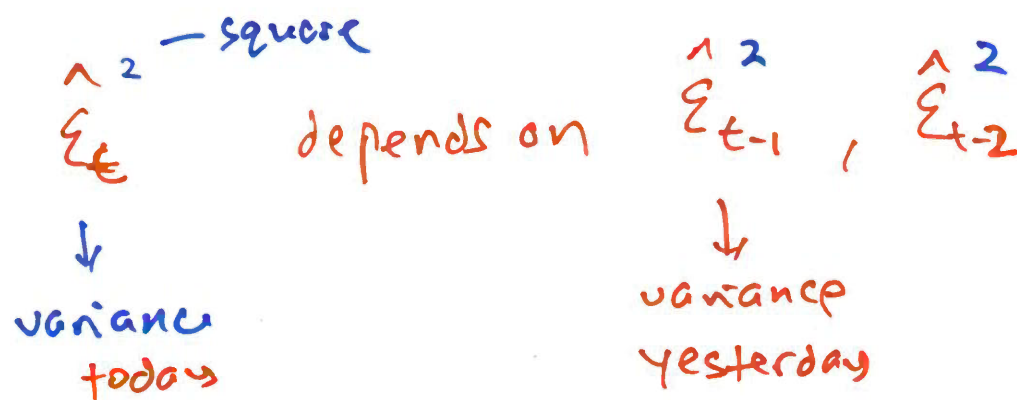
Broadly speaking, the volatility of an asset can be measured by the **standard deviation** of its returns.

Volatility is preferred to variance because it is measured in the same units as returns.

We are interested in **conditional standard deviations** associated with relatively “low”- frequency returns, such as daily or monthly frequency.

Testing for time varying volatility

- ▶ How do we know if we need a time varying conditional volatility model?
→ becomes "variance"
- ▶ Plots of $\hat{\varepsilon}_t^2$ (usually also seen in r_t^2) against time suggest time varying conditional variance (which implies time varying volatility).
- ▶ Autocorrelations for $\hat{\varepsilon}_t^2$ or r_t^2 suggest a conditional variance (and hence volatility) that depends on its past.
- ▶ In practice, we see autocorrelations in r_t^2 , even when we don't see autocorrelations in r_t .

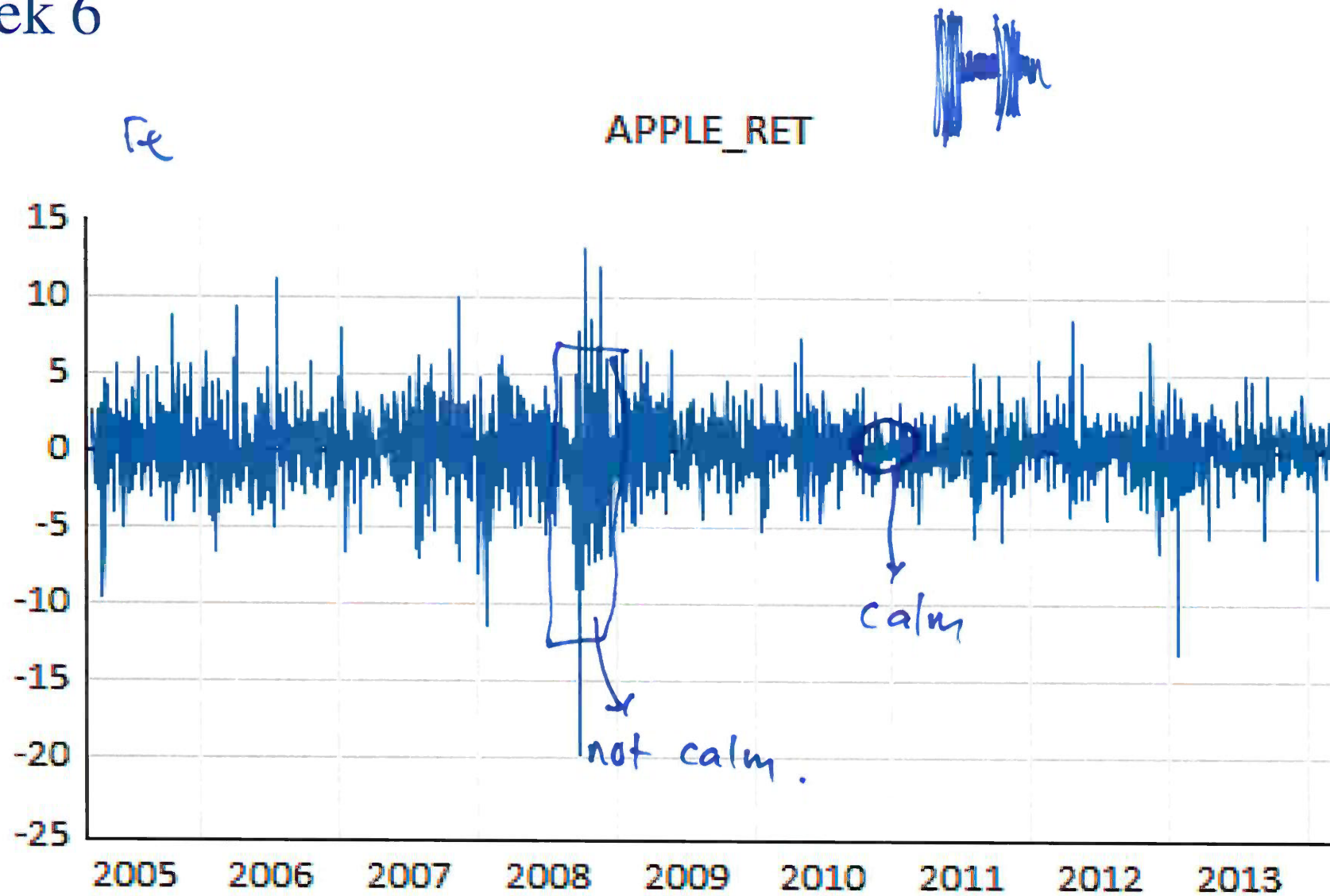


IBM log return

Consider the daily prices for the U.S. Apple stock from 14 March 2005 to 13 March 2014 (Week 6).

EMH

Week 6



Correlogram (Week 6)

```
> source("correlogram.R")
```

```
> length(lr)
```

```
[1] 2348
```

```
> correlogram(lr, max.lag = 10)
```

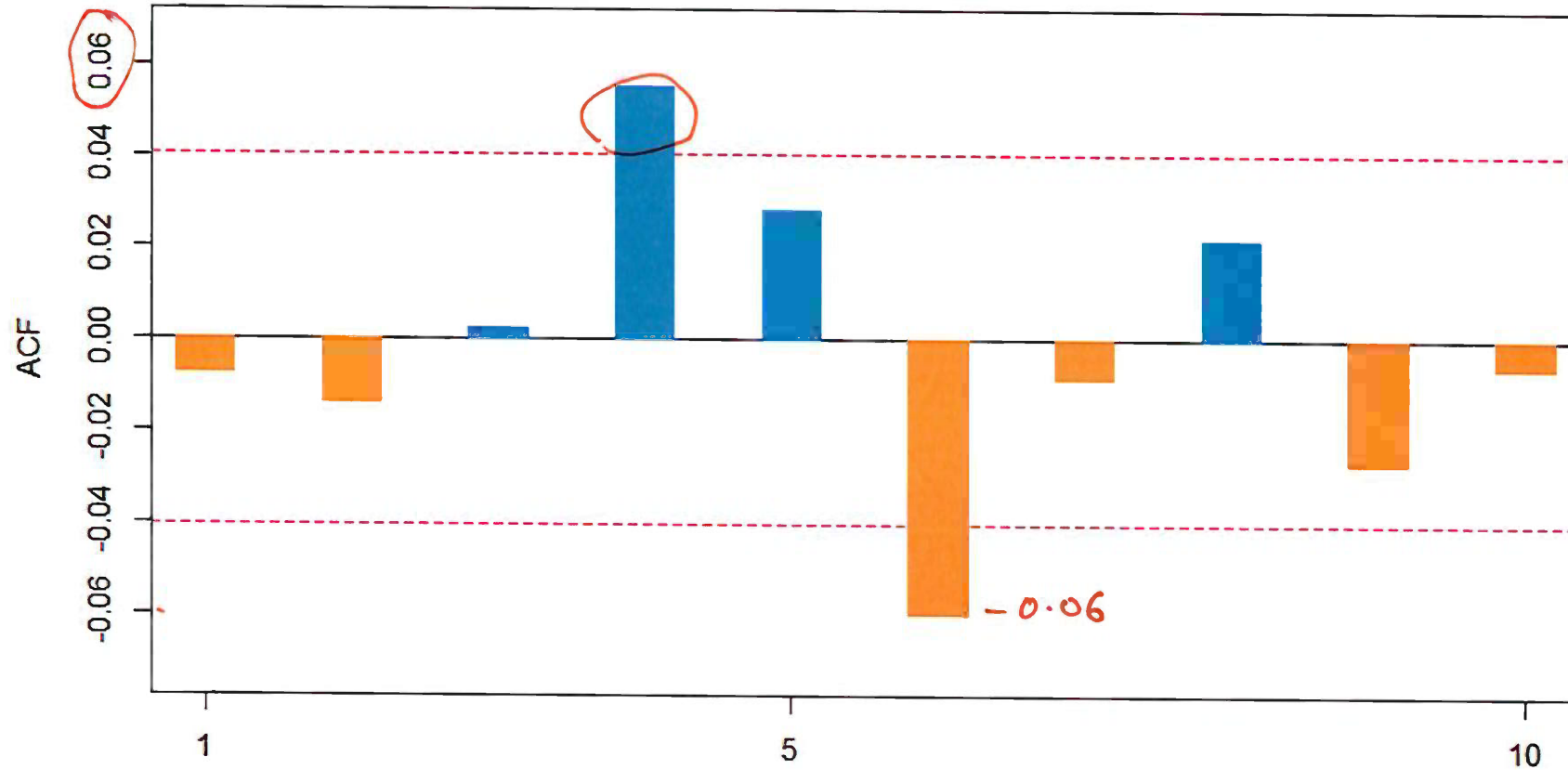
Lag	AC	Q	Prob
1	-0.0072	0.1230	0.725800
2	-0.0137	0.5619	0.755100
3	0.0025	0.5770	0.901700
4	0.0553	7.7662	0.100500
5	0.0284	9.6641	0.085330
6	-0.0600	18.1523	0.005863
7	-0.0085	18.3245	0.010590
8	0.0218	19.4402	0.012670
9	-0.0271	21.1726	0.011910
10	-0.0061	21.2604	0.019350

ACF

$\gamma(k)$ for $k = 1, 2, 3,$

r_t

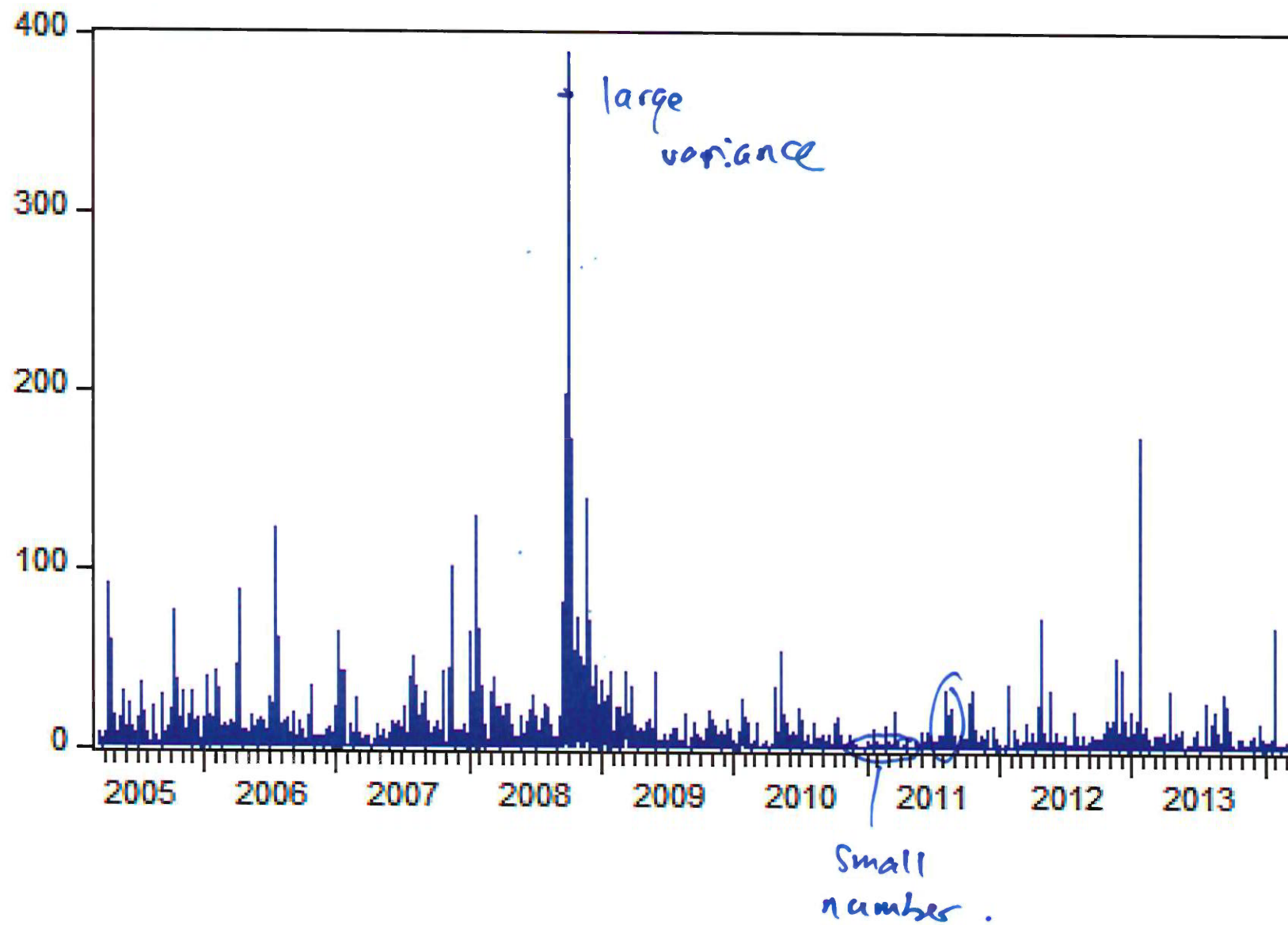
Week 6 ACF (Correlogram)



"almost" serially uncorrelated

"white noise."

Squared log returns r_t^2



Correlogram of Squared log returns

```
> lr_sq = lr^2
```

```
> plot(lr_sq, main = "Squared Log returns")
```

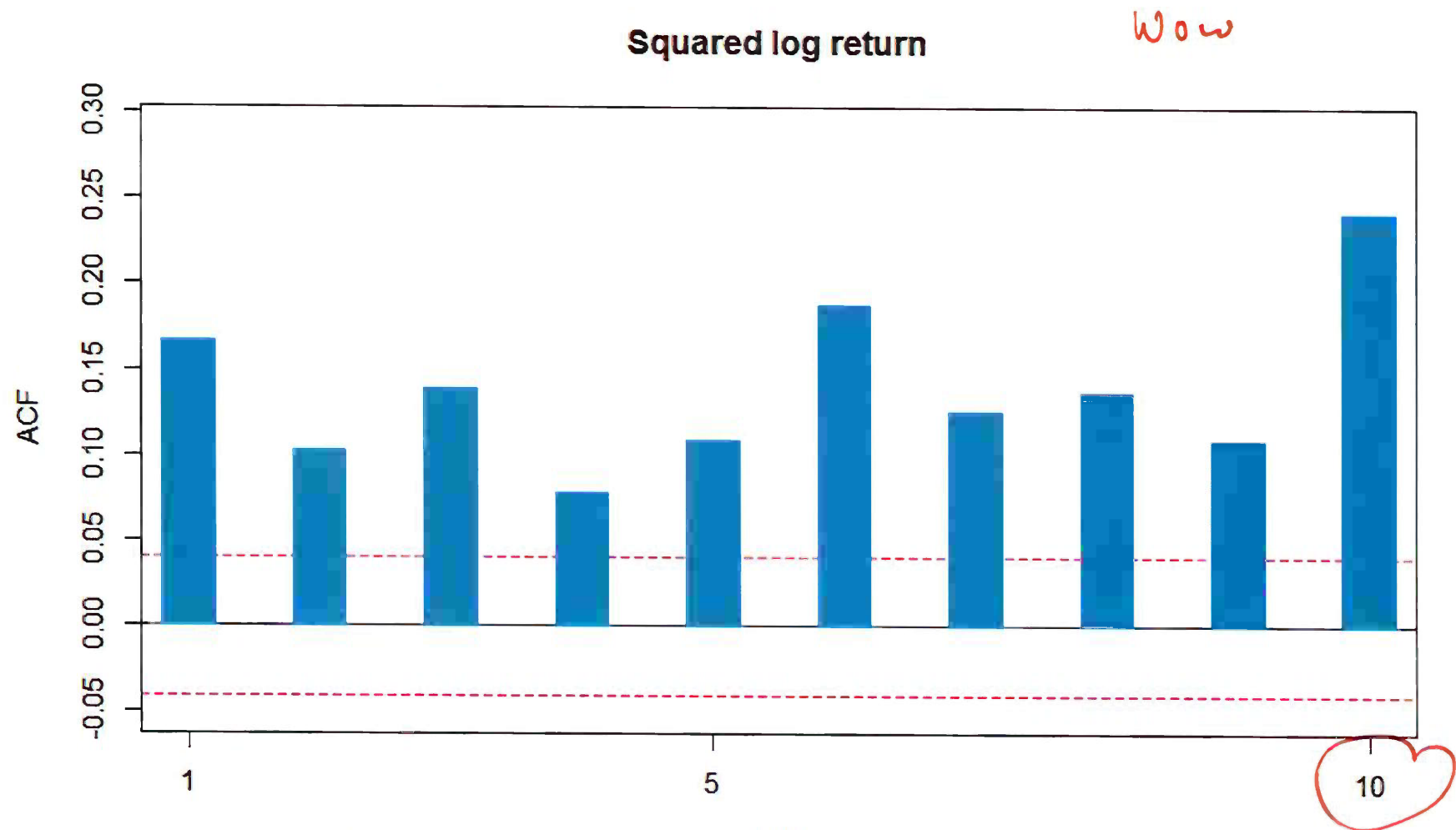
```
> correlogram(lr_sq, max.lag = 10, main.acf = "Squared log ret
```

Lag	AC	Q	Prob
1	0.1668	65.4198	6.053e-16
2	0.1032	90.4529	2.282e-20
3	0.1386	135.6721	3.240e-29
4	0.0779	149.9444	2.092e-31
5	0.1090	177.9123	1.494e-36
6	0.1878	261.0172	1.810e-53
7	0.1257	298.2342	1.441e-60
8	0.1364	342.1258	4.338e-69
9	0.1087	370.0246	3.373e-74
10	0.2409	507.0068	1.404e-102

Slide 29

↓
no zeros.

Correlogram of r_t^2



$\hat{\epsilon}_t^2$ depends on $\hat{\epsilon}_{t-1}^2, \hat{\epsilon}_{t-2}^2, \hat{\epsilon}_{t-3}^2, \dots$

Testing for time varying volatility

- For a returns model such as

$$r_t = E[r_t | \mathcal{F}_{t-1}] + \varepsilon_t$$

- if there is serial correlation (autocorrelation) in ε_t^2 (and hence r_t^2)
 $\implies$ a constant volatility model will be inadequate.
- We can test for this serial correlation:
 - with a rejection of " $H_0 : \varepsilon_t^2$ are serially uncorrelated" suggesting some kind of AR model for the conditional variance that predicts ε_t^2 or $\mathbb{V}(\varepsilon_t | \mathcal{F}_{t-1})$ using ε_{t-k}^2 .
 - This structure is called **autoregressive conditional heteroskedasticity** (i.e. ARCH) and tests of " $H_0 : \varepsilon_t^2$ are serially uncorrelated" are called test for ARCH effects.

week 6
except "square"

week 6

ε_t depends on $\varepsilon_{t-1}, \varepsilon_{t-2}, \dots$

↓

ε_t is autocorrelated $E(\varepsilon_t \varepsilon_{t-k}) \neq 0 \quad k=1, 2, 3, \dots$

ARCH test

If the true model is

$$y_t = x_t' \beta + \varepsilon_t, \quad \text{with} \quad \varepsilon_t^2 = \rho_0 + \rho_1 \varepsilon_{t-1}^2 + u_t,$$

but we estimate

$$y_t = x_t' \beta + \varepsilon_t,$$

$$\begin{array}{l} \uparrow \\ H_0 : \rho_1 = 0 \Rightarrow \text{ARCH} \\ \downarrow \\ \text{constant variance.} \end{array}$$

then there will be a dynamic pattern remaining in the **squared** residuals.

$$H_A : \rho_1 \neq 0$$

likely to Reject H_0 .

ARCH test

To test $H_0 : \rho_1 = 0$ (against $H_1 : \rho_1 \neq 0$)

1. We first estimate the conditional mean equation $y_t = x_t' \beta + \varepsilon_t$.
(we'll use an AR(p) model for this).
2. Then obtain square the residuals $\hat{\varepsilon}_t^2$, and run the auxilliary regression $\hat{\varepsilon}_t^2 = \gamma_0 + \gamma_1 \hat{\varepsilon}_{t-1}^2 + u_t$.
3. The LM test statistic TR^2 is distributed as a χ_1^2 under H_0 .
4. If TR^2 is greater than the critical value from χ_1^2 distribution, then we reject H_0 .

ARCH test

- If the true model is

$$y_t = x_t' \beta + \varepsilon_t, \quad \text{with} \quad \varepsilon_t^2 = \rho_0 + \rho_1 \varepsilon_{t-1}^2 + \rho_2 \varepsilon_{t-2}^2 \dots + \rho_q \varepsilon_{t-q}^2 + u_t,$$

but we estimate

AR(2)
$$y_t = x_t' \beta + \varepsilon_t,$$

then there will be a dynamic pattern remaining in the squared residuals.

- To test $H_0 : \rho_1 = \rho_2 = \dots = \rho_q = 0$ (against $H_1 : \text{at least one } \rho_k \neq 0, \text{ for } k = 1, 2, \dots, q$) we run

$$\hat{\varepsilon}_t^2 = \rho_0 + \rho_1 \hat{\varepsilon}_{t-1}^2 + \rho_2 \hat{\varepsilon}_{t-2}^2 + \dots + \rho_q \hat{\varepsilon}_{t-q}^2 + u_t.$$

- Under H_0 , the $TR^2 \sim \chi_q^2$ and we will reject the null if TR^2 is 'large'.
- A rejection of H_0 suggests that volatility is time varying and predictable.

ARCH test

An example: IBM log returns. We fit an AR(2) model and then analyse the residuals.

> `fit_ar2 <- arima(as.numeric(lr), order = c(2,0,0),`
`+ include.mean = TRUE)`

> `fit_ar2`

call:

`arima(x = as.numeric(lr), order = c(2, 0, 0), include.mean = TRUE)`

coefficients:

	ar1	ar2	intercept
$\hat{\phi}_1$	-0.0073	-0.0137	0.1114
s.e.	0.0206	0.0206	0.0455

AR



$\hat{\phi}_2$

$$\hat{\mu} = E(\hat{Y}_t) = \frac{1}{1 - \hat{\phi}_1 - \hat{\phi}_2}$$

unconditional mean

sigma^2 estimated as 5.069: log likelihood = -5237.29, aic = 10482.57

> `res_ar2 = residuals(fit_ar2)` $\Rightarrow$ gives $\hat{\varepsilon}_t$

$$Y_t = \hat{C} + \hat{\phi}_1 Y_{t-1} + \hat{\phi}_2 Y_{t-2} + \hat{\varepsilon}_t$$

↓
not this

> res_ar2 = residuals(fit_ar2) $\hat{\epsilon}_t$ not square

> **FintS::ArchTest**(res_ar2, lags = 2) slide 40

→ will square ϵ_t for you

ARCH LM-test; Null hypothesis: no ARCH effects $H_0: \rho_1 = 0$

data: res_ar2

Chi-squared = 78.435, df = 2, p-value < 2.2e-16

test = LM = ARCH test = 78.435

p-value = 0

Reject H_0 for sure.

ARCH test with 2-lag

We estimated an AR(2) model

$$r_t - 0.114 = -0.0073 (r_{t-1} - 0.114) - 0.0137 (r_{t-2} - 0.114) + \hat{\varepsilon}_t$$

$\hat{\mu} \Rightarrow$ intercept $\hat{\phi}_1$ $\hat{\phi}_2$ de-mean.
 $\hat{\mu}$ $AR(1) - \hat{\mu}$ $AR(2) - \hat{\mu}$

and the auxiliary regression

$$\hat{\varepsilon}_t^2 = \rho_0 + \rho_1 \hat{\varepsilon}_{t-1}^2 + \rho_2 \hat{\varepsilon}_{t-2}^2$$

To test

$$H_0 : \rho_1 = \rho_2 = 0 \quad \text{No ARCH effects}$$

$$H_1 : \text{at least one } \rho_k \neq 0 \text{ for } k = 1, 2$$

The LM test statistic $TR^2 = 78.435$ with $p\text{-value} = 0$. We reject the null hypothesis and suggests that volatility is time varying and predictable.

1:09

1:14 back

ARCH Models

Consider the following model

$$r_t = c + \phi_1 r_{t-1} + \varepsilon_t$$

with

$$\varepsilon_t = u_t \sigma_t$$

u_t is new this "error term"

where

$$\mathbb{E}(u_t | \mathcal{F}_{t-1}) = 0 \quad \text{and} \quad \mathbb{E}(u_t^2 | \mathcal{F}_{t-1}) = 1.$$

mds

unit variance.

1. Show that $E(\varepsilon_t | \mathcal{F}_{t-1}) = 0$.

usual

2. Show that $\mathbb{V}(\varepsilon_t | \mathcal{F}_{t-1}) = \sigma_t^2$

↑

ε_t usual → to show

u_t new "error term." → Assume.

New assumption

whaaaaa t
1. Because σ_t is known given $\mathcal{F}_{t-1}$

$$\varepsilon_t = u_t \sigma_t \rightarrow \text{focus.}$$

$$E(\varepsilon_t | \mathcal{F}_{t-1}) = E(u_t \sigma_t | \mathcal{F}_{t-1}) = \sigma_t \underbrace{E(u_t | \mathcal{F}_{t-1})}_{\text{m.d.s.}} = 0.$$

2.

$$\begin{aligned} \mathbb{V}(\varepsilon_t | \mathcal{F}_{t-1}) &= E(\varepsilon_t^2 | \mathcal{F}_{t-1}) \\ &= E(u_t^2 \sigma_t^2 | \mathcal{F}_{t-1}) \\ &= \sigma_t^2 E(u_t^2 | \mathcal{F}_{t-1}) \quad \text{because } \sigma_t^2 \text{ is known given } \mathcal{F}_{t-1} \\ &\quad \text{1 a unit variance.} \\ &= \sigma_t^2 \\ &\quad \downarrow \\ &\text{time-dependent.} \end{aligned}$$

independently and identically distributed (i.i.d.)

u_t is assumed to be independently and identically distributed for all t (i.i.d.), that is

$$u_t \sim \text{i.i.d. } (0, \sigma^2).$$

This notation means that u_t has mean zero and variance σ^2 .

The assumption that u_t is independent over time means that it has no autocorrelation at any lag. $\Rightarrow$ like u_t is a white noise.

It also has a constant variance σ^2 over time so it is homoskedastic.

A time series that has no autocorrelation and is homoskedastic is sometimes called *white noise*. Therefore u_t is an example of a white noise process.

ARCH models

Return to our modelling framework in which

$$r_t = E[r_t | \mathcal{F}_{t-1}] + \varepsilon_t$$

e.g. $E[r_t | \mathcal{F}_{t-1}] = c + \phi_1 r_{t-1}$.

ARCH models specify

new
↑
 $\varepsilon_t = u_t \sigma_t,$

where

$$u_t \sim i.i.d.(0, 1), \quad \begin{array}{l} E(u_t) = 0 \\ \text{var}(u_t) = 1 \end{array}$$

and

$$\sigma_t^2 = \mathbb{V}(\varepsilon_t | \mathcal{F}_{t-1}) = g(\varepsilon_{t-1}^2, \varepsilon_{t-2}^2, \dots).$$

Such models are referred to as being **conditionally heteroskedastic**.

↓
 $g(\cdot)$ function.

ARCH models

The errors of the mean specification $\{\varepsilon_t\}$ are often referred to as the **‘news process’**.

The news process $\{\varepsilon_t\}$ is driven by underlying **shocks** u_t , and it is common to assume that $u_t \sim i.i.d.N(0, 1)$.

This assumption implies that

$$\varepsilon_t = u_t \sigma_t$$

$$u_t = \frac{\varepsilon_t}{\sigma_t} \stackrel{iid}{\sim} N(0, 1),$$

and

$$\varepsilon_t | \mathcal{F}_{t-1} \sim N(0, \sigma_t^2)$$

$$\text{var}(\varepsilon_t) = \sigma^2$$

Here, ε_t is only **conditionally normal**.

$$\begin{aligned} E(\varepsilon_t | \mathcal{F}_{t-1}) &= 0 && \text{slide 42} \\ V(\varepsilon_t | \mathcal{F}_{t-1}) &= \sigma_t^2 && \text{" "} \end{aligned}$$

ARCH models

a unit variance .



$u_t = \frac{\varepsilon_t}{\sigma_t}$ is often called the **standardised news**.

$\frac{\widehat{\varepsilon}_t}{\widehat{\sigma}_t}$ is the **standardised residual**.

ARCH models

For simplicity, assume that $\mathbb{E}[r_t | \mathcal{F}_{t-1}] = 0$.

We have:

$$\begin{aligned} r_t &= \varepsilon_t \\ &= u_t \sigma_t \\ &\equiv u_t \sqrt{g(\varepsilon_{t-1}^2, \varepsilon_{t-2}^2, \dots, \varepsilon_{t-k}^2)}. \end{aligned}$$

ARCH models then depend on:

1. Functional form of $g(\cdot)$;
2. Information in $g(\cdot)$, i.e., lags of ε_{t-k}^2 .

ARCH(1) model:

is known given $\mathcal{F}_{t-1}$ slide 43

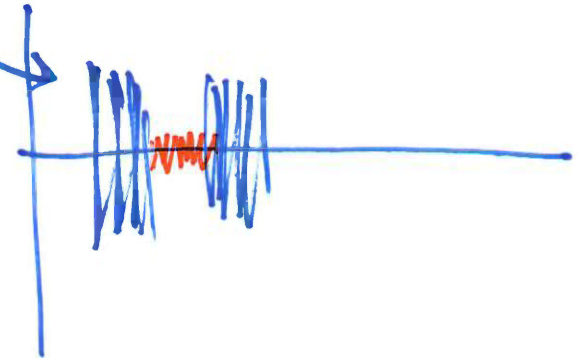
slide 42

$$\sigma_t^2 = \mathbb{V}(\varepsilon_t | \mathcal{F}_{t-1}) = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 \Rightarrow \sigma_t^2 = 10 \text{ or } 20$$

70 70

This model generates **volatility clustering**:

- ▶ large errors tend to be followed by large errors
- ▶ small errors tend to be followed by small errors



We assume that $\alpha_i > 0$, for $i = 0, 1$ to ensure that $\sigma_t^2 > 0$.

We also assume $\alpha_1 < 1$ to ensure the unconditional variance is well defined.

Common ARCH model

An ARCH(q) model assumes that $E(\varepsilon_t | \mathcal{F}_{t-1}) = 0$ and

$$\sigma_t^2 = \mathbb{V}(\varepsilon_t | \mathcal{F}_{t-1}) = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \alpha_2 \varepsilon_{t-2}^2 + \dots + \alpha_q \varepsilon_{t-q}^2.$$

We assume that the $\alpha_i > 0$, for $i = 1, 2, \dots, q$, to ensure that $\sigma_t^2 > 0$.

We also assume $\sum_{i=1}^q \alpha_i < 1$ to ensure the unconditional variance is well defined.

Common ARCH model

Recall that ARCH(q) model

hetero $\leftarrow \sigma_t^2 = \mathbb{V}(\varepsilon_t | \mathcal{F}_{t-1}) = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \alpha_2 \varepsilon_{t-2}^2 + \dots + \alpha_q \varepsilon_{t-q}^2.$

conditional on $\mathcal{F}_{t-1}$

AR $\Rightarrow$ lags of "q"

The α parameters determine how past news affect current volatility (persistence).

Persistence is often measured by $\sum_{i=1}^q \alpha_i$.

If $\alpha_1 = \dots = \alpha_q = 0$, then $\sigma_t^2 = \alpha_0$ and the model reverts to a standard constant volatility model.

no ARCH.

Consider an AR(1)-ARCH(1) model

$$r_t = c + \phi_1 r_{t-1} + \varepsilon_t \quad \text{AR(1)}$$

with

$$\varepsilon_t = u_t \sigma_t$$

where $u_t \sim i.i.d.(0, 1)$ and

$$\sigma_t^2 = \mathbb{V}(\varepsilon_t | \mathcal{F}_{t-1}) = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 \quad \text{ARCH(1)}$$

ARCH stands for Autoregressive Conditional Heteroskedasticity.

ARCH and Stylized Facts

Recall our stylized facts:

1. Heavy tails (positive excess kurtosis).
2. Asymmetry (negative skewness).
3. Lack of (weak) persistent in levels of returns.
4. Volatility clustering.

Unconditional moments of the ARCH(1) model

To understand which of the stylized facts ARCH(1) satisfies, we must analyze the unconditional moments of $\{\varepsilon_t\}$.

We can work out several facts, based on the ARCH(1) specification

$$\varepsilon_t = \sigma_t u_t$$

↓ new error

$$u_t \stackrel{iid}{\sim} N(0, 1)$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2.$$

Unconditional moments of the ARCH(1) model

A key tool for this analysis is the Law of Iterated Expectations (L.I.E.):

$$\mathbb{E}[X_t] = \mathbb{E}[\mathbb{E}[X_t | \mathcal{F}_{t-1}]],$$

and the Law of Total Variance:

$$\mathbb{V}[X_t] = \mathbb{E}[\mathbb{V}[X_t | \mathcal{F}_{t-1}]] + \mathbb{V}[\mathbb{E}[X_t | \mathcal{F}_{t-1}]].$$

Unconditional moments of the ARCH(1) model

The L.I.E. says that we can find unconditional expectations of random variables by first taking conditional expectations.

The Law of Total Variance says we can find unconditional variance using conditional variance and conditional expectation.

Unconditional moments of the ARCH(1) model

(1.) The **unconditional mean** ARCH(1) error is zero

$$E(\varepsilon_t) = 0.$$

$\varepsilon_t \Rightarrow$ usual

$$\varepsilon_t = \sigma_t u_t$$

(2.) For ARCH(1) errors, $Cov(\varepsilon_t, \varepsilon_{t-k}) = 0$ for all $k \geq 1$.

(3.) The **unconditional variance** of the ARCH(1) error is

$$\mathbb{V}(\varepsilon_t) = \frac{\alpha_0}{(1 - \alpha_1)} \cdot \begin{aligned} &= \text{a number.} \\ &= LR \text{ variance.} \end{aligned}$$

Unconditional moments of the ARCH(1) model

(4.) If we assume $u_t \stackrel{iid}{\sim} N(0, 1)$, then $E(\varepsilon_t^3) = 0$.

(5.) It can be shown that the kurtosis for ε_t is

$$\kappa = 3 \times \frac{(1 - \alpha_1^2)}{(1 - 3\alpha_1^2)} > 3.$$

when $0 < \alpha_1 < \sqrt{1/3}$ (proof not supplied).

Unconditional moments of the ARCH(1) model

(1.) The **unconditional mean** ARCH(1) error is zero

$$E(\varepsilon_t) = 0.$$

Proof.

By the L.I.E.,

$$E(\varepsilon_t) = E[E(\varepsilon_t|\mathcal{F}_{t-1})],$$

and since $\varepsilon_t|\mathcal{F}_{t-1} \sim N(0, \sigma_t^2)$ it follows that $E(\varepsilon_t|\mathcal{F}_{t-1}) = 0$.

Therefore $E(\varepsilon_t) = E(0) = 0$. □

Unconditional moments of the ARCH(1) model

(2.) For ARCH(1) errors, $Cov(\varepsilon_t, \varepsilon_{t-k}) = 0$ for all $k \geq 1$.

Proof.

By definition of covariance (and stationarity of $\{\varepsilon_t\}$)

$$\begin{aligned} Cov(\varepsilon_t, \varepsilon_{t-k}) &= E[(\varepsilon_t - E(\varepsilon_t))(\varepsilon_{t-k} - E(\varepsilon_{t-k}))] \\ &= E[\varepsilon_t \varepsilon_{t-k}] \end{aligned}$$

since, by (1), $E(\varepsilon_t) = E(\varepsilon_{t-k}) = 0$. Further, the L.I.E. implies

$$E[\varepsilon_t \varepsilon_{t-k}] = E[E[\varepsilon_t \varepsilon_{t-k} | \mathcal{F}_{t-1}]] = E[\varepsilon_{t-k} E[\varepsilon_t | \mathcal{F}_{t-1}]],$$

since, given $\mathcal{F}_{t-1}$, the error ε_{t-k} is known if $k \geq 1$. Then, since $E[\varepsilon_t | \mathcal{F}_{t-1}] = 0$,

$$E[\varepsilon_t \varepsilon_{t-k}] = E[\varepsilon_{t-k} \times 0] = 0.$$

□

This shows that an ARCH(1) process is not autocorrelated.

Unconditional moments of the ARCH(1) model

(3.) The **unconditional variance** of the ARCH(1) error is

$$\mathbb{V}(\varepsilon_t) = \frac{\alpha_0}{(1 - \alpha_1)}.$$

Proof.

First note that $E(\varepsilon_t) = 0$ implies that $\mathbb{V}(\varepsilon_t) = E(\varepsilon_t^2)$. By the L.I.E.,

$$E(\varepsilon_t^2) = E[E(\varepsilon_t^2 | \mathcal{F}_{t-1})].$$

Next, note that since $\varepsilon_t | \mathcal{F}_{t-1} \sim N(0, \sigma_t^2)$ and $E(\varepsilon_t | \mathcal{F}_{t-1}) = 0$, then

$$\sigma_t^2 = \mathbb{V}(\varepsilon_t | \mathcal{F}_{t-1}) = E(\varepsilon_t^2 | \mathcal{F}_{t-1}).$$

(Continued on next slide)



Unconditional moments of the ARCH(1) model

Proof.

(continued) Hence

$$\begin{aligned}\mathbb{V}(\varepsilon_t) &= E(\varepsilon_t^2) = E[E(\varepsilon_t^2 | \mathcal{F}_{t-1})] \\ &= E(\sigma_t^2) = \alpha_0 + \alpha_1 E(\varepsilon_{t-1}^2) \\ &= \alpha_0 + \alpha_1 \mathbb{V}(\varepsilon_{t-1}).\end{aligned}$$

But if ε_t is stationary, then

$$\mathbb{V}(\varepsilon_{t-1}) = \mathbb{V}(\varepsilon_t),$$

and so

$$\mathbb{V}(\varepsilon_t) = \alpha_0 + \alpha_1 \mathbb{V}(\varepsilon_t).$$

Solving this, we get the result that $\mathbb{V}(\varepsilon_t) = \alpha_0 / (1 - \alpha_1)$. □

Unconditional moments of the ARCH(1) model

(4.) If we assume $u_t \stackrel{iid}{\sim} N(0, 1)$, then $E(\varepsilon_t^3) = 0$.

Proof.

Note that by the L.I.E.,

$$E(\varepsilon_t^3) = E[E[\varepsilon_t^3 | \mathcal{F}_{t-1}]].$$

Writing $\varepsilon_t = \sigma_t u_t$, we have

$$\begin{aligned} E(\varepsilon_t^3) &= E[E[\sigma_t^3 u_t^3 | \mathcal{F}_{t-1}]] \\ &= E[\sigma_t^3 E[u_t^3 | \mathcal{F}_{t-1}]], \end{aligned}$$

since σ_t is a function of past information only. Then, since by assumption $u_t \sim N(0, 1)$, we have $E[u_t^3 | \mathcal{F}_{t-1}] = 0$ and so $E(\varepsilon_t^3) = E[\sigma_t^3 \times 0] = 0$. □

This shows that the unconditional distribution of the errors is symmetric.